New lower bounds for multicolour van der Waerden numbers from composite cyclic moduli

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October 8, 2026
Produced autonomously by Claude (Opus 5.5) agents in Claude Code; run and submitted by Aleksandr Bersenev.
Not an official Anthropic publication. Not peer reviewed.

Abstract

Let $W(r,k)$ be the least N such that every r-colouring of $\{1,\dots,N\}$ contains a monochromatic k-term arithmetic progression. We give explicit colourings showing $W(4,6) > 255{,}606$, $W(5,6) > 1{,}278{,}030$, $W(6,6) > 2{,}531{,}178$, $W(4,8) > 18{,}874{,}790$, $W(4,9) > 212{,}844{,}870$, $W(5,10) > 13{,}086{,}046{,}358$, $W(4,11) > 3{,}761{,}291{,}754$, $W(4,12) > 37{,}088{,}321{,}982$ and seven further bounds for five and six colours (Table 1). To our knowledge, as of October 8, 2026, these improve fifteen entries of the standard tables (the Wikipedia table of $W(r,k)$ and Monroe's table of four-colour bounds). No new search is involved. The colourings come from one elementary blow-up lemma applied to cyclic 2- and 3-colourings that are already in the literature (Rabung; Herwig–Heule–van Lambalgen–van Maaren; Rabung–Lotts; Heule; cf. Kouril–Paul for $W(2,6)$). These certificates are cyclic on *composite* moduli (226, 1642, 5158, 19394, 58066) that exceed the prime moduli used in the published product bounds. We also use the Blankenship–Cummings–Taranchuk block step with the prime $p = 11 > k$ for $k = 10$, as their own remark allows. Every bound is certified by a kernel-checked proof in Lean 4 (core only, no Mathlib) whose main theorems depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

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1 Introduction

For integers $r \geq 1$ and $k \geq 2$, van der Waerden's theorem guarantees a least integer $W(r,k)$ such that every colouring of $\{1,\dots,W(r,k)\}$ with r colours contains a monochromatic k-term arithmetic progression (AP). We use the order of arguments of the Wikipedia table [15]: r colours and k-term APs. A *lower bound* "$W(r,k) > N$" means that some r-colouring of N consecutive integers has no monochromatic k-AP; the published tables write it as "$> N$".

Only a handful of non-trivial values are known exactly, among them $W(2,6) = 1132$ (Kouril–Paul [9]) and $W(3,4) = 293$ (Kouril). Every other entry rests on an explicit colouring. The main sources of record colourings are the following.

- *Power-residue colourings* of $\mathbb{Z}/p$ (Rabung [12]).
- *Cyclic zippers*, which double or quadruple such a colouring to a colouring of $\mathbb{Z}/2p$ or $\mathbb{Z}/4p$ (Herwig, Heule, van Lambalgen and van Maaren [5]; Rabung and Lotts [13]).
- *SAT searches*, sometimes with symmetry constraints (see Heule [4] for the history, and Komkov [7]).
- *Recursive constructions* for many colours:
  - Xu's product recursion $W(k,st) > p(W(k,t)-1)+1$ for primes p admitting an s-colouring of $\mathbb{Z}/p$ without k-APs, as described in Monroe [10, 11] (we only use the form $W(st,k)-1 \geq n(W(t,k)-1)$ of Theorem 3.2, without the final $+1$, which is not reproduced by Theorem 3.1);
  - the block recursion $W(r,k) > p \cdot (W(r-\lceil r/p \rceil, k)-1)$ of Blankenship, Cummings and Taranchuk (BCT) [1], where p is the largest prime $\leq k$.

Monroe's distributed computation [10, 11] produced the current four-colour column (e.g. $W(4,6) >$ 157,348, $W(4,8) >$ 12,288,155). For $k \geq 7$ the five- and six-colour columns of the Wikipedia table are obtained from it by the BCT recursion (Monroe's own tables stop at four colours); for $k = 6$ the entries are $W(5,6) > 786{,}740 = 5 \cdot 157{,}348$ (BCT) and $W(6,6) > 1{,}555{,}549 = 139 \cdot 11{,}191$ (Xu's product of the prime 139 with $W(3,6) > 11{,}191$). Monroe states explicitly that the "ring van der Waerden number" in Xu's recursion is taken over the prime field $\mathbb{F}_p$, so the moduli in those products are primes: 139, 1069, 3389, 11497, 17863, 58013. Asymptotic lower bounds have advanced recently (Hunter [6]; Fox–Hunter [2]; Campos–Fox–Schildkraut [14]), but these papers do not tabulate small values.

What is new here. A periodic colouring is just as good as a colouring of a prime field, provided its period is fully cyclically AP-free. Several record two-colour certificates are of this form, with composite periods larger than the primes used. Combining them with a general blow-up lemma (Lemma 3.1) gives Theorem 1.1. The lemma contains both the product recursion and the BCT block recursion. We stress that the contribution is modest and combinatorially elementary.

- The underlying cyclic colourings were found by others.
- The product construction is Xu's; Hunter [6, Lemma 3.2] states it for direct products of abelian groups.
- The BCT block step with $p > k$ is remarked in [1].

What appears not to have been done is to apply these constructions with the best available cyclic moduli and to record the resulting numbers. We also give a complete formal certificate.

Theorem 1.1. *The lower bounds in the column "new" of Table 1 hold. Each is witnessed by an explicit colouring of* $\{0, \dots, N-1\}$ *described in Section 4, and each is proved in Lean 4 (Section 5).*

entry	previous best (source)	**new**	construction
$W(4,6)$	157,348 [11] (157,209 in [15])	**255,606**	$226 \cdot 1131$
$W(5,6)$	786,740 [15]	**1,278,030**	$5 \cdot 255{,}606$
$W(6,6)$	1,555,549 [15]	**2,531,178**	$2238 \cdot 1131$
$W(4,8)$	12,288,155 [11, 15]	**18,874,790**	$1642 \cdot 11{,}495$
$W(5,8)$	86,017,085 [15]	**132,123,530**	$7 \cdot W(4,8)$-bound
$W(6,8)$	602,119,595 [15]	**924,864,710**	$7 \cdot W(5,8)$-bound
$W(4,9)$	139,847,085 [11, 15]	**212,844,870**	$5158 \cdot 41{,}265$
$W(5,9)$	978,929,595 [15]	**1,489,914,090**	$7 \cdot W(4,9)$-bound
$W(6,9)$	6,852,507,165 [15]	**10,429,398,630**	$7 \cdot W(5,9)$-bound
$W(5,10)$	8,327,484,046 [15]	**13,086,046,358**	$11 \cdot 1{,}189{,}640{,}578$
$W(6,10)$	58,292,388,322 [15]	**91,602,324,506**	$7 \cdot W(5,10)$-bound
$W(4,11)$	3,464,368,083 [11, 15]	**3,761,291,754**	$19394 \cdot 193{,}941$
$W(5,11)$	38,108,048,913 [15]	**41,374,209,294**	$11 \cdot W(4,11)$-bound
$W(6,11)$	419,188,538,043 [15]	**455,116,302,234**	$11 \cdot W(5,11)$-bound
$W(4,12)$	37,054,469,451 [11]	**37,088,321,982**	$58066 \cdot 638{,}727$

Table 1: Lower bounds $W(r,k) > N$. All entries mean ">". "Previous best" is the Wikipedia table (wikitext retrieved October 8, 2026) and Monroe's four-colour table (arXiv v7, Nov. 2022; journal version, Dec. 2025). The Wikipedia table stops at $k = 11$, so the last row is compared with Monroe only. In the last column, "$p \cdot X$-bound" means p times the new bound in the row named.

Along the way the Lean development also certifies several known values: $W(2,6) > 1131$, $W(2,8) > 11{,}495$, $W(2,9) > 41{,}265$, $W(2,10) > 103{,}474$, $W(2,11) > 193{,}941$, $W(2,12) > 638{,}727$ and Monroe's $W(4,10) > 1{,}189{,}640{,}578$. None of these is new. The rows $k = 3, 4, 5, 7$ of the table are not affected by our method (Section 6).

2 Definitions

Throughout, colourings are maps $c : \mathbb{N} \to \mathbb{N}$, and only their values on the relevant interval matter. We work with $\{0, \dots, N-1\}$ instead of $\{1, \dots, N\}$; shifting by one changes nothing.

Definition 2.1. Let $k \geq 2$. A colouring c has *no monochromatic k-AP in* $[0, N)$ if for all $a \geq 0$ and $d \geq 1$ with $a + (k-1)d < N$ there is an $i < k$ with $c(a + id) \neq c(a)$. We write $W(r,k) > N$ if some c with $c([0,N)) \subseteq \{0, \dots, r-1\}$ has this property.

Definition 2.2. Let $n \geq 1$. A colouring χ of $\mathbb{Z}/n = \{0, \dots, n-1\}$ is *cyclically k-AP-free* if for every $a \in \mathbb{Z}/n$ and every $d \not\equiv 0 \pmod{n}$ the residues $a, a+d, \dots, a+(k-1)d \pmod{n}$ do not all have the same colour. Here d may have small additive order. In that case the "progression" repeats elements, and the condition still applies.

Lemma 2.3 (unrolling)**.** *Let $k \geq 3$ and $n \geq 1$. A colouring c of $[0, (k-1)n)$ with period n has no monochromatic k-AP if and only if the induced colouring χ of $\mathbb{Z}/n$ is cyclically k-AP-free.*

Proof. A k-AP in $[0,(k-1)n)$ with difference d satisfies $(k-1)d<(k-1)n$, so $d<n$ and its residues form a cyclic k-AP with difference $d\not\equiv 0$. Conversely, take a cyclic k-AP (a,d). Reversing the order if necessary, we may assume $1\le d\le n/2$ and $0\le a<n$. Then $a+(k-1)d\le n-1+(k-1)n/2\le (k-1)n-1$ because $k\ge 3$, so the progression occurs inside $[0,(k-1)n)$. □

Hence a certificate of the form "$k-1$ periods of length n" is the same thing as a cyclically AP-free colouring of $\mathbb{Z}/n$. The zipper certificates of [5, 13] and the certificates in Heule's repository [3] are of this form. (The files `W_2_6_226`, `W_2_8_1642`, `W_2_9_5158` and `W_3_6_2238` of [3] contain exactly the $(k-1)n$ symbols of the $k-1$ periods; the additional element that gives the length $(k-1)n+1$ is found by the search described in Section 4.)

3 The blow-up lemma

Lemma 3.1 (blow-up)**.** *Let $n,t\ge 1$ and $F:\mathbb{Z}/n\times\{0,\dots,t-1\}\to\mathbb{N}$. Assume:*

(i) $F(z,\cdot)$ is injective for every z;

(ii) for every value γ, the set $P_\gamma=\{z\in\mathbb{Z}/n:\ \gamma=F(z,u)$ for some $u<t\}$ contains no cyclic k-AP $\{a+id \bmod n : i<k\}$ with $d\not\equiv 0 \pmod n$.

Let $\psi:[0,M)\to\{0,\dots,t-1\}$ have no monochromatic k-AP. Then

$$c(x)=F\big(x \bmod n,\ \psi(\lfloor x/n\rfloor)\big)$$

has no monochromatic k-AP in $[0,nM)$. In particular, if F takes at most r values, then $W(r,k)-1\ \ge\ n\,(W(t,k)-1)$.

Proof. Suppose $a+id$ ($i<k$, $d\ge 1$) lies in $[0,nM)$ and is monochromatic of colour γ.

If $n\mid d$, write $d=nq$ with $q\ge 1$. All terms have the residue $z=a \bmod n$, and $\lfloor (a+id)/n\rfloor=\lfloor a/n\rfloor+iq$. These indices form a k-AP in $[0,M)$, since $n(\lfloor a/n\rfloor+(k-1)q)\le a+(k-1)d<nM$. From $F(z,\psi(\lfloor a/n\rfloor+iq))=\gamma$ for all i and condition (i), the values $\psi(\lfloor a/n\rfloor+iq)$ are all equal, which contradicts the hypothesis on ψ.

If $n\nmid d$, the residues $(a+id) \bmod n=((a \bmod n)+i(d \bmod n)) \bmod n$ form a cyclic k-AP with difference $d \bmod n\neq 0$. Each term has colour $\gamma=F(z_i,u_i)$ with $u_i<t$, so the progression lies in P_γ, which contradicts (ii). □

We call such an F an (r,t)-*cover* of $\mathbb{Z}/n$ (for k-APs). Three special cases give all our bounds.

Corollary 3.2 (products; Xu, cf. [11, 6])**.** *If χ is a cyclically k-AP-free s-colouring of $\mathbb{Z}/n$, then $W(st,k)-1\ge n(W(t,k)-1)$.*

Proof. Take $F(z,u)=t\chi(z)+u$. Then $P_\gamma=\chi^{-1}(\lfloor\gamma/t\rfloor)$. □

Corollary 3.3 (blocks; cf. BCT [1])**.** *Let p be prime and let $2\le r\le p$. Put $F(z,u)=(z \bmod r+1+u) \bmod r$ for $u<r-1$, so that colour $z \bmod r$ is the one colour* missing *at z. Assume one of the following:*

(a) $p\le k$;

(b) $p>k$ and every residue class of $\{0,\dots,p-1\}$ modulo r has at least $p-k+1$ elements.

Then $W(r,k)-1\ge p\,(W(r-1,k)-1)$.

Proof. Here $P_\gamma = \{z \in \mathbb{Z}/p : z \not\equiv \gamma \pmod r\}$. Since p is prime, a cyclic k-AP in $\mathbb{Z}/p$ with $d \neq 0$ has $\min(p, k)$ distinct elements, and if $p \leq k$ it is all of $\mathbb{Z}/p$. In case (a), $P_\gamma \neq \mathbb{Z}/p$, because every residue class modulo r meets $\{0, \dots, p-1\}$ when $r \leq p$. In case (b), $|P_\gamma| \leq k-1$. In both cases P_γ contains no cyclic k-AP. Injectivity in u holds because $u \mapsto z \bmod r + 1 + u$ takes $r - 1 < r$ consecutive values. □

Case (a) with p the largest prime $\leq k$ is the BCT step for $r \leq p$. Case (b) is the strengthening noted at the end of [1, §2] ("choosing a larger prime p such that $p - \lfloor p/r \rfloor < k$"). For $k = 10$ and $r = 5$ it allows $p = 11$: the classes modulo 5 in $\{0, \dots, 10\}$ have $3, 2, 2, 2, 2 \geq 2$ elements.

Corollary 3.4 (extra element)**.** *If c has no monochromatic k-AP in $[0, N)$, and no k-AP ending at N is monochromatic, then c has none in $[0, N]$.*

This trivial statement is used to realise $W(2, k) - 1 = (k-1)n + 1$: we take a suitable rotation of the unrolled cyclic colouring and append one element, which is checked directly.

4 The constructions

4.1 Cyclic colourings used

Table 2 lists the cyclic colourings (files in `data/`, with SHA-256 hashes in `data/SHA256SUMS`). The files contain one digit per element of $\mathbb{Z}/n$.

k	n	col.	origin	file, sha256 prefix
6	226	2	zipped power-residue colouring mod 113 [5], which reproduces the SAT bound $W(2,6) > 1131$ of Kouril–Franco [8] and is extremal since $W(2,6) = 1132$ [9]; file `W_2_6_226` of [3], rotated by 112	`chi_6_226`, `2c720049`
6	2238	3	Heule's certificate `W_3_6_2238` [3, 4]	`chi3_6_2238`, `aad85c14`
8	1642	2	zipped mod 821 [5]; file `W_2_8_1642` of [3], rotated by 820	`chi_8_1642`, `2eaf8f5e`
9	5158	2	zipped mod 2579 [5]; file `W_2_9_5158` of [3], rotated by 2578	`chi_9_5158`, `a91414a7`
10	11497	2	Rabung quadratic-residue colouring mod 11497 [12]	`chi_10_11497`, `29dd34cc`
11	19394	2	zipped mod 9697 [13] (our reconstruction)	`chi_11_19394`, `7d4c4e82`
12	58066	2	zipped mod 29033 [13] (our reconstruction)	`chi_12_58066`, `73a1e273`

Table 2: Cyclically k-AP-free colourings of $\mathbb{Z}/n$ (files in `data/`).

Reconstructions. For $p = 11497$ we take $\chi(z) = \mathrm{ind}_g(z) \bmod 2$ for $z \neq 0$, where g is the least primitive root modulo p, and set $\chi(0) = 0$. For the zipped colourings modulo $2p$ with $p = 9697$ and $p = 29033$, we let q be the quadratic-residue colouring modulo p and put

$$\chi(x) = q(x \bmod p) \oplus (x \bmod 2) \quad \text{for } p \nmid x, \qquad \chi(0) = 0, \quad \chi(p) = 1.$$

Unrolling these and appending one element reproduces the published bounds $W(2, 10) > 103{,}474$, $W(2, 11) > 193{,}941$ and $W(2, 12) > 638{,}727$ of Rabung and Rabung–Lotts. For every row of Table 2 the cyclic property was checked exhaustively by a C program (all a, all $d \leq n/2$) and independently in the Lean kernel (Section 5).

4.2 Derivation of Table 1

Let $L(r,k)$ denote the length of the best colouring used; thus $W(r,k) > L(r,k)$.

- *Two colours.* For (k,n) from Table 2, $L(2,k) = (k-1)n+1$: the unrolled colouring rotated by s_0, followed by one element of colour e. The values of (s_0,e) for $k = 6,8,9,10,11,12$ are $(0,0)$, $(821,0)$, $(2579,0)$, $(0,1)$, $(9697,0)$, $(29033,0)$. They were found by exhaustive search and are recorded in `Main.lean`.
- *Four colours* (Corollary 3.2, $s = t = 2$): $L(4,k) = n \cdot L(2,k)$. This gives $226 \cdot 1131$, $1642 \cdot 11{,}495$, $5158 \cdot 41{,}265$, $11497 \cdot 103{,}474$ (Monroe's value), $19394 \cdot 193{,}941$ and $58066 \cdot 638{,}727$.
- *Six colours, $k = 6$* (Corollary 3.2, $s = 3$, $t = 2$): $L(6,6) = 2238 \cdot 1131$.
- *Five colours* (Corollary 3.3, $r = 5$): $L(5,6) = 5\,L(4,6)$ ($p = 5 \le k$); $L(5,k) = 7\,L(4,k)$ for $k = 8,9$; $L(5,10) = 11\,L(4,10)$ ($p = 11 > k$, case (b)); $L(5,11) = 11\,L(4,11)$.
- *Six colours, $k \ge 8$* (Corollary 3.3, $r = 6$): $L(6,k) = 7\,L(5,k)$ for $k = 8,9,10$ and $L(6,11) = 11\,L(5,11)$.

The resulting colourings are explicit. For example, the four-colour colouring witnessing $W(4,6) > 255{,}606$ is $c(x) = 2\chi_{226}(x \bmod 226) + \psi_6(\lfloor x/226 \rfloor)$, where $\psi_6(y) = \chi_{226}(y \bmod 226)$ for $y < 1130$ and $\psi_6(1130) = 0$.

Remark 4.1. Monroe's products use the primes 139, 1069, 3389, 17863 and 58013 for $k = 6,8,9,11,12$ (for $k = 8,\dots,12$ the primes are listed in [10, 11, Table 2]; for $k = 6$ the prime 139, whose quadratic-residue colouring is cyclically 6-AP-free, is inferred from the Wikipedia value $157{,}209 = 139 \cdot 1131$). Monroe's own entry $W(4,6) > 157{,}348$ equals $139 \cdot 1132$, i.e. $139 \cdot W(2,6)$ rather than $139 \cdot (W(2,6)-1)$, and we could not reconstruct a colouring of that length; the point is immaterial here, since 255,606 exceeds both values, but it also affects the Wikipedia entry $W(5,6) > 786{,}740 = 5 \cdot 157{,}348$. The improvement for $k = 6,8,9$ is a factor 226/139, 1642/1069, 5158/3389, roughly 1.5 to 1.6. For $k = 11$ and $k = 12$ it is 8.6% and 0.09%. For $k = 10$ the four-colour bound is unchanged, and the gain in five colours comes only from $p = 11$ in place of $p = 7$. Other entries propagate through the block step.

5 Formal verification

All bounds of Table 1 are proved in a single self-contained Lean file (`lean/Main.lean`, 591 KB). We used Lean 4.34.1 with core and `Std` only and no Mathlib. The two definitions used in the statements are the first declarations of the file:

```
def NoMonoAP (k N : Nat) (c : Nat → Nat) : Prop :=
  ∀ a d : Nat, 0 < d → a + (k - 1) * d < N → ∃ i, i < k ∧ c (a + i * d) ≠ c a

def VdWGt (r k N : Nat) : Prop :=
  ∃ c : Nat → Nat, (∀ x, x < N → c x < r) ∧ NoMonoAP k N c
```

The main theorems are, verbatim (proof terms omitted):

```
theorem W4_6  : VdWGt 4 6 255606
theorem W5_6  : VdWGt 5 6 1278030
theorem W6_6  : VdWGt 6 6 2531178
```

```
theorem W4_8  : VdWGt 4 8 18874790
theorem W5_8  : VdWGt 5 8 132123530
theorem W6_8  : VdWGt 6 8 924864710
theorem W4_9  : VdWGt 4 9 212844870
theorem W5_9  : VdWGt 5 9 1489914090
theorem W6_9  : VdWGt 6 9 10429398630
theorem W5_10 : VdWGt 5 10 13086046358
theorem W6_10 : VdWGt 6 10 91602324506
theorem W4_11 : VdWGt 4 11 3761291754
theorem W5_11 : VdWGt 5 11 41374209294
theorem W6_11 : VdWGt 6 11 455116302234
theorem W4_12 : VdWGt 4 12 37088321982
```

The file also contains `W2_6`, ..., `W2_12` and `W4_10` (known values).

Statement audit. `NoMonoAP k N c` quantifies over all a and all $d \geq 1$ whose last term $a+(k-1)d$ lies in $[0, N)$, that is, over all k-APs in $\{0, \ldots, N-1\}$. Its conclusion, "some term has a colour different from the first", is exactly "not monochromatic". `VdWGt r k N` also requires all colours on $[0, N)$ to be $< r$. Shifting by one gives a colouring of $\{1, \ldots, N\}$, so `VdWGt r k N` is equivalent to $W(r, k) > N$, the same ">" convention as the tables. The arguments are (colours, length, N), so `W4_6` is the four-colour, six-term statement. The main theorems have no hypotheses, and all numerals are literals.

Proof architecture.

- *General lemmas, proved by hand.* The blow-up lemma (`blowup`, Lemma 3.1); `product` (Corollary 3.2); `unroll` and `unrollN` (one direction of Lemma 2.3); `cover_step` (Corollary 3.3, with the cover condition checked by `decide`); and `NoMonoAP.extend_one` (Corollary 3.4).
- *Finite facts.* These are checked by `decide +kernel` on Boolean checkers with proved soundness lemmas.
- *Two-colour cyclic colourings.* Each is invariant under $z \mapsto mz$ for a unit m generating an index-2 subgroup of units: $m = 9, 5, 3, 2, 3, 9$ for $n = 226$, 1642, 5158, 11497, 19394, 58066. The kernel checks three things: this invariance; a table of "steps to an orbit representative" for the differences d; and all start points a for the at most five representatives. The representatives are, e.g., $\{1, 2, 3, 6, 113\}$ for $n = 226$ and $\{1, 5\}$ for $n = 11497$. Lemma `cyc_of_symP` then transfers the check to all differences, by induction on the number of steps.
- *The 3-colouring of* $\mathbb{Z}/2238$. It has no multiplicative symmetry, so it is checked for all a and all $d \leq 1119$. A reversal lemma (`cyc_of_rows`) covers $d > n/2$.
- *Memory.* Large checks are split into chunks; this keeps the kernel memory bounded.

Axioms and run. Every main theorem prints

```
'W4_6' depends on axioms: [propext, Classical.choice, Quot.sound]
```

and likewise for all 22 theorems listed in `lean/lean_output.txt`. There is no `sorryAx` and no `native_decide` (`Lean.ofReduceBool`). The complete file checked with no errors and no warnings in 2310 s of wall time, with peak resident memory 3.6 GB. The command was `nice -n 10 lean`

`-j1 Main.lean` (Lean 4.34.1), run under a timeout and an RSS watchdog; the authors' guarded wrapper script could not be used because its address-space limit made Lean fail at start-up (`lean/leanchk_attempt.txt`). The SHA-256 of the checked file is `20251d5c...f4f`. An independent re-run of `lean Main.lean` with the same toolchain on October 8, 2026 reproduced the output verbatim (exit status 0, 1237 s wall time, peak RSS 3.5 GiB; log in `lean/lean_output_review.txt`).

Independent checks. Separately from Lean, C programs (not included in this distribution; the Lean file is the certificate) performed the following checks:

- every colouring of Table 2 was checked exhaustively;
- every two-colour interval colouring $L(2,k)$ was brute-forced over all APs;
- the colouring for $W(4,6) > 255{,}606$ was brute-forced over all 6-APs, with no monochromatic one;
- each of the larger final colourings was tested on $2 \cdot 10^7$ random APs, with no monochromatic one, and two deliberately wrong constructions were correctly rejected.

6 Limitations and open questions

- *Novelty.* The comparison is with the Wikipedia table (retrieved October 8, 2026) and Monroe's tables [10, 11]. Our searches of arXiv and the web (2023–2026) found no later improvement of these entries. A bound in an unindexed source cannot be excluded; hence "to our knowledge, as of October 8, 2026".
- *Nature of the contribution.* No new search result enters Theorem 1.1. The gains come from combining published certificates with known recursions. We searched for genuinely new (r,t)-covers with SAT (CaDiCaL) and exhaustive enumeration, and found none beyond the counting ones of Corollary 3.3. Specifically, within the ranges of n that we enumerated (and disregarding the trivial moduli $n < k$, for which every proper subset of $\mathbb{Z}/n$ is cyclically k-AP-free):
  - $(5,3)$-covers for 4-APs exist only for $n \le 5$ (an independent exhaustive search confirms this for all $n \le 16$);
  - $(6,4)$-covers exist only for $n = 5$ (for 5-APs) and $n \le 7$ (for 6-APs); for 6-APs, maximal 6-AP-free subsets of $\mathbb{Z}/11$ and $\mathbb{Z}/13$ have density $< 2/3$;
  - no $(6,3)$-cover of $\mathbb{Z}/n$ for 5-APs was found for $45 \le n \le 53$. Such a cover with $n \ge 46$ would give $W(6,5) > 46 \cdot 2173 = 99{,}958$ and thus beat Komkov's bound $W(6,5) > 99{,}554$ [7] (the Wikipedia entry $> 98{,}748$ predates it).

  The rows $k = 3, 4, 5, 7$ are therefore unchanged.
- *Open.* Two questions remain. Are there cyclically AP-free 2-colourings of $\mathbb{Z}/n$ with $(k-1)n+1$ beyond the current $W(2,k)$ records? For $k = 7$ and $k = 10$ the best moduli are the primes 617 and 11497. Do fractional covers (an (r,t)-cover with r/t not an integer, or with $r/t = s$ but not coming from an s-colouring) exist for moduli larger than the best s-colourings? Either would propagate through Lemma 3.1.

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